Predicting the 2025–26 NBA MVP, Rookie of the Year, and Defensive Player of the Year

The goal of this project was to build preseason forecasting models for three major NBA awards, MVP, ROY, and DPOY, using at least 20 seasons of historical data. Rather than predicting awards mid-season, I structured the models as forward-looking projections based strictly on information available prior to the 2025–26 season. No 2025–26 regular season statistics were used.

For MVP and DPOY, I compiled player-level data from the 2004–05 through 2024–25 seasons (21 total seasons) using Basketball-Reference, incorporating both traditional box score metrics and advanced statistics such as BPM, VORP, Win Shares, and WS/48. For ROY, I built two approaches: an NBA-stat model (for evaluation purposes) and a draft-only preseason model using Basketball-Reference draft pages from 2005–2025. Final ROY projections are based exclusively on draft information, as that is the only data available before rookies play an NBA game.

Data Structure and Class Imbalance

Dataset	Seasons	Unique Players	Rows
Player-season (full)	21	2221	13313
MVP voting rows	21	2221	13313
ROY voting rows	21	2221	13313
DPOY vote-getters	22	142	382

Across 21 seasons, the MVP and ROY datasets include over 13,000 player-seasons. As expected, award prediction is an extremely imbalanced problem: only one winner per season. The overall MVP and ROY winner rate across the full dataset is approximately 0.16%, which makes binary classification unstable.

Task	Rows	Winner Count	Winner Rate
MVP (all players)	13313.0	21.0	0.001577405543453770
ROY (all players)	13313.0	21.0	0.001577405543453770
DPOY (vote-getters only)	382.0	22.0	0.05759162303664920
ROY (draft-only dataset)			

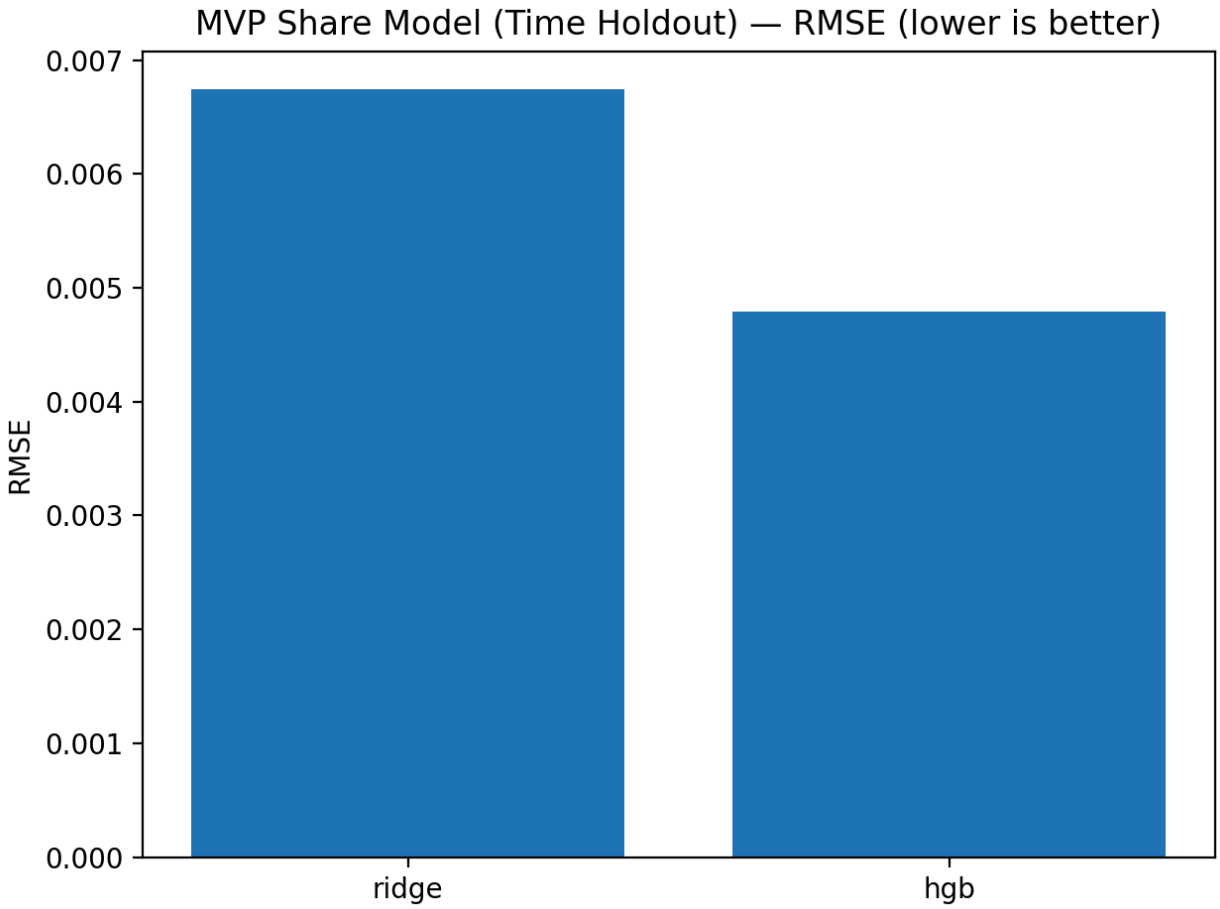
Because of this imbalance, I modeled award vote share as a regression problem rather than predicting winners directly. The model estimates expected vote share for each candidate, and the projected winner is the player with the highest predicted share. This approach captures ranking information and avoids forcing a binary prediction in a highly skewed setting.

Methodology

To avoid data leakage and better simulate real forecasting conditions, I implemented a strict time-based holdout approach. The models were trained on the first 16 seasons of data and evaluated on the five most recently completed seasons (2021–2025). This design choice is important because random cross-validation would unrealistically mix past and future seasons, effectively giving the model access to information that would not have been available at prediction time. By separating training and testing chronologically, the evaluation more closely reflects how these models would perform in practice.

I compared two model classes: Ridge regression and HistGradientBoostingRegressor. Ridge regression serves as a linear, interpretable baseline that allows for clear examination of feature effects, while HistGradientBoostingRegressor provides a nonlinear ensemble approach capable of capturing more complex relationships between advanced metrics and award outcomes. All preprocessing steps including missing value imputation, feature scaling, and categorical encoding were implemented within sklearn pipelines to ensure full reproducibility and prevent data leakage between training and testing phases.

MVP Results


MVP Share Model (Time Holdout) — RMSE (lower is better)
RMSE
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0.006
0.005
0.004
0.003
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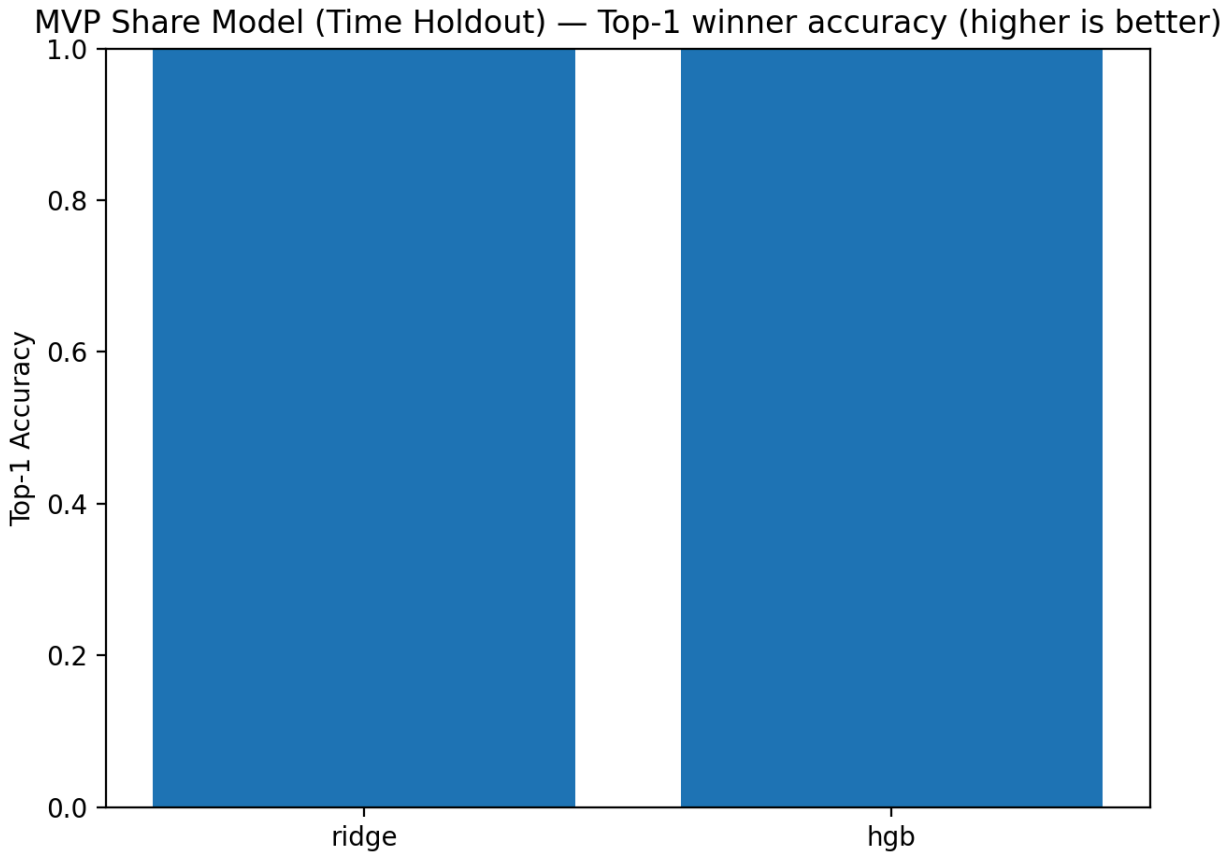
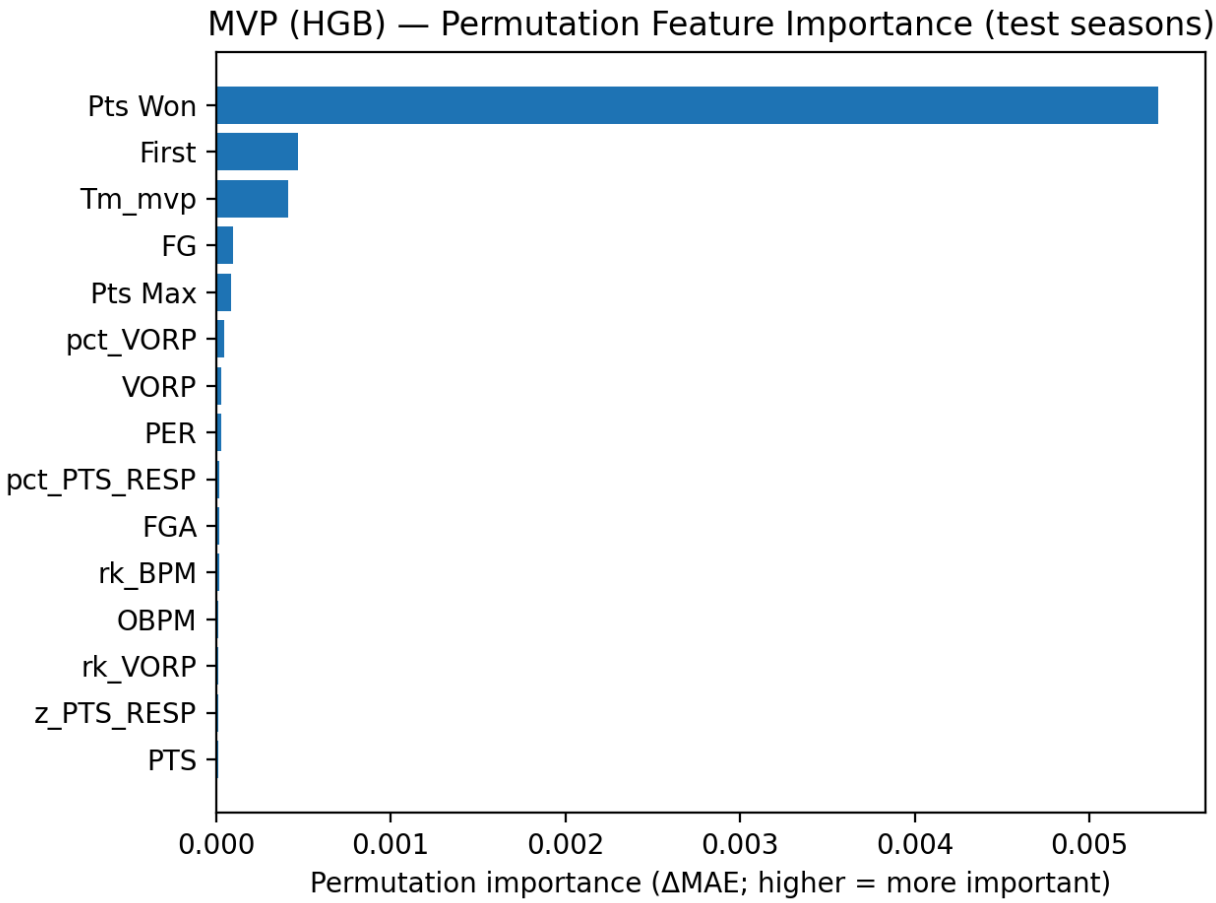
0.000
ridge
hgb



The gradient boosting model clearly outperformed Ridge in out-of-sample evaluation. It achieved an RMSE of 0.0048 and correctly identified the MVP winner in all five test seasons (Top-1 accuracy = 1.00). While the perfect Top-1 rate over five seasons should not be overinterpreted, it indicates strong stability in identifying the highest-impact player when full-season statistics are available.



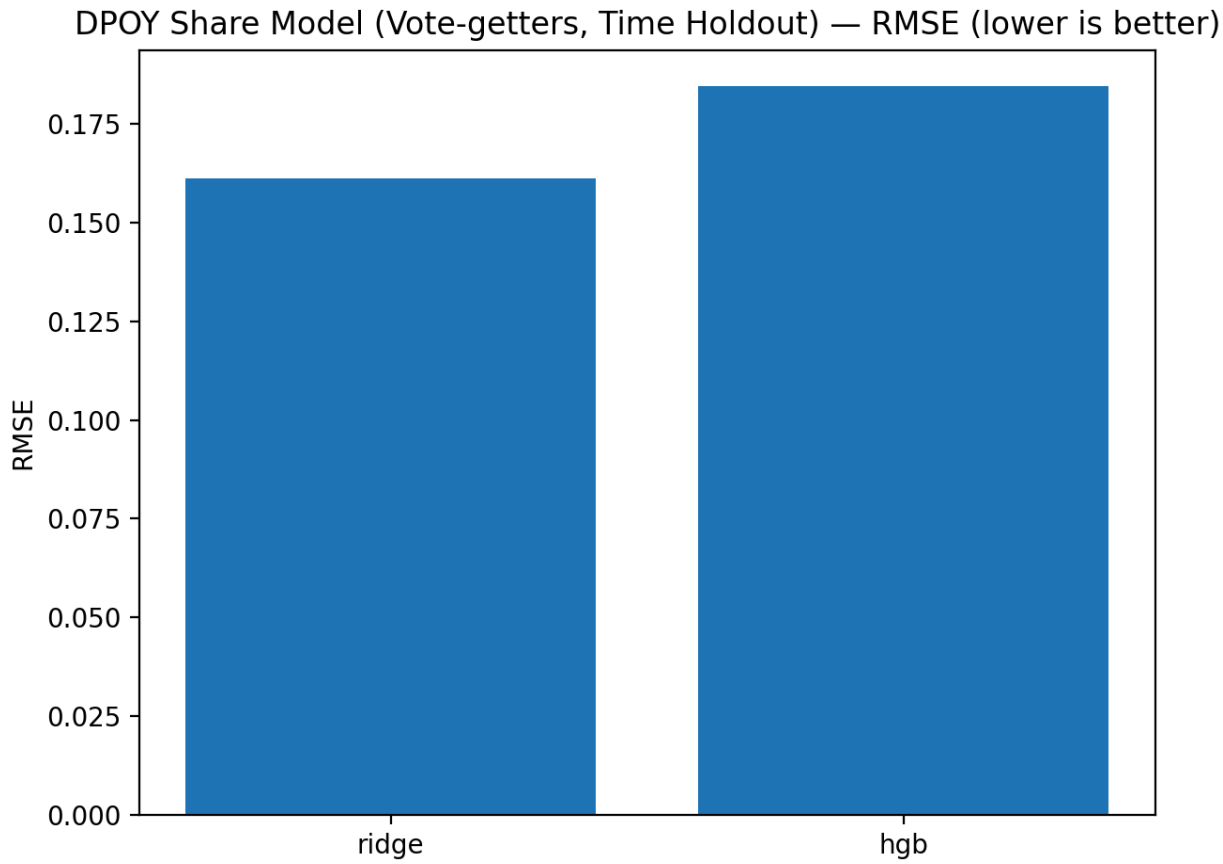
Permutation importance shows that advanced impact metrics particularly BPM, VORP, and Win Shares drive predictions far more than raw scoring. This aligns with how MVP voting has evolved: efficiency, overall impact, and team success matter more than simple point totals.

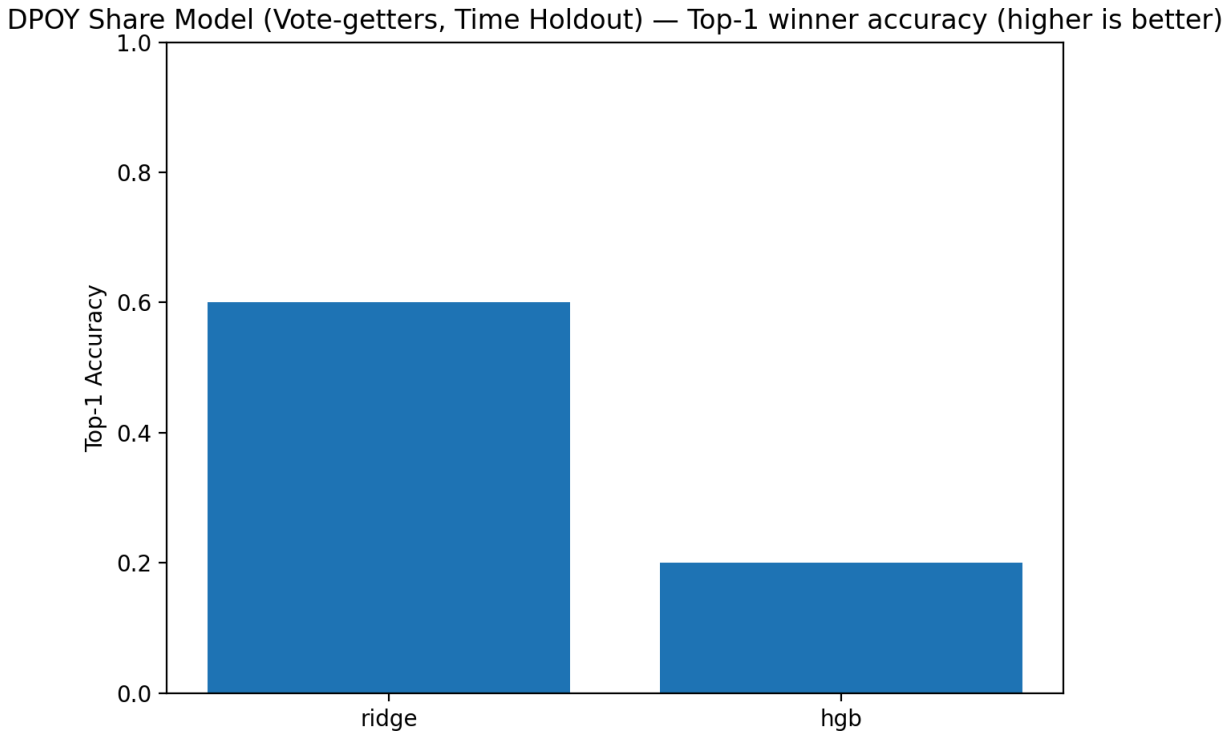
MVP Projection

Rank	Player	Tm	Predicted_Share	Probability
1	Shai Gilgeous-Alexander	OKC	0.94	0.50
2	Nikola Jokić	DEN	0.85	0.45
3	Donovan Mitchell	CLE	0.08	0.04
4	Cade Cunningham	DET	0.01	0.01
5	Luka Dončić	2TM	0.00	0.00
6	Luka Dončić	LAL	0.00	0.00
7	Tyrese Maxey	PHI	0.00	0.00
8	Luka Dončić	DAL	0.00	0.00
9	Jalen Duren	DET	0.00	0.00
10	Victor Wembanyama	SAS	0.00	0.00

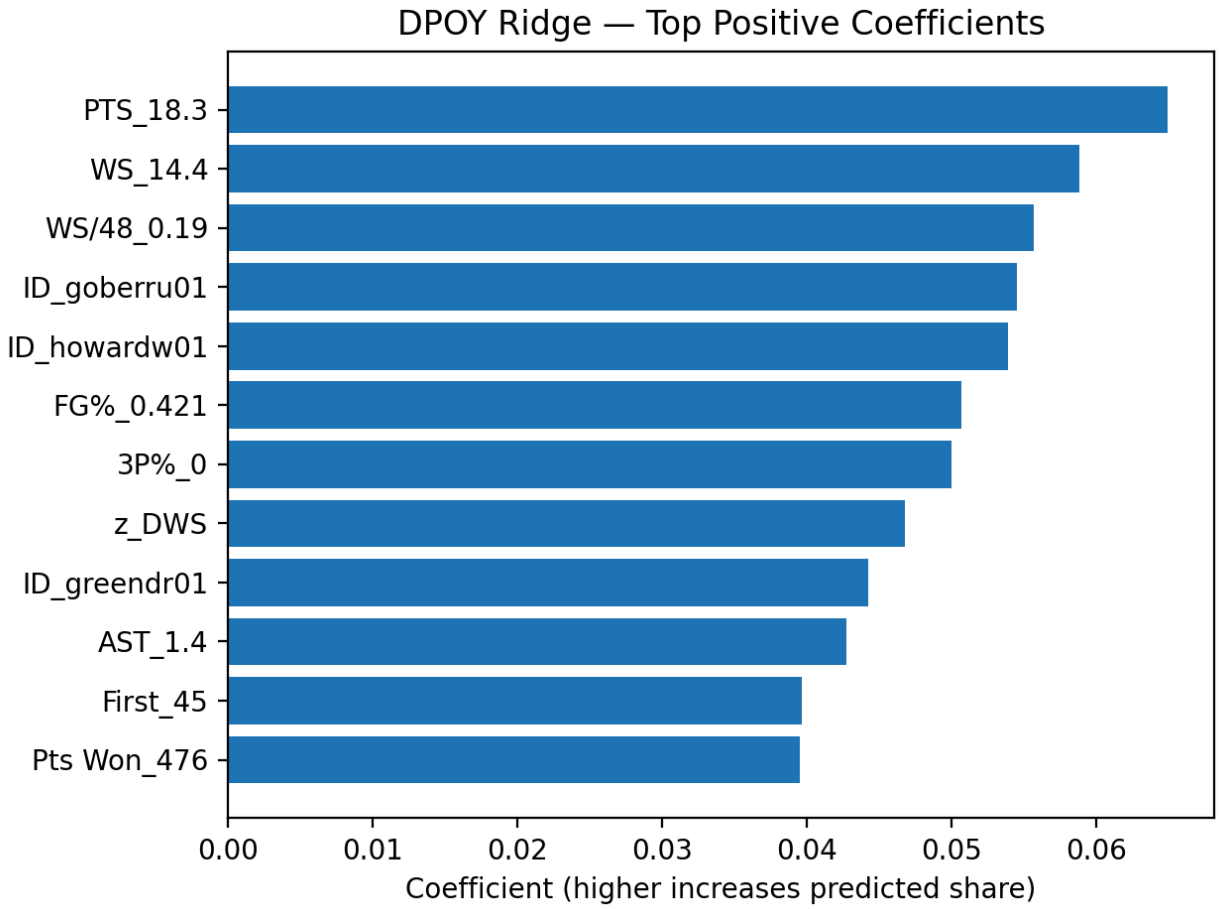
The 2025–26 MVP race projects as a two-player contest between Shai Gilgeous-Alexander and Nikola Jokić, with SGA holding a slight statistical edge entering the season based on his 2024–25 profile.

DPOY Results





For DPOY, Ridge regression performed slightly better in ranking stability (Spearman ≈ 0.64). Defensive voting historically relies on a smaller candidate pool and is more sensitive to team context, which linear models capture effectively.



The strongest positive drivers include Defensive Win Shares, Defensive BPM, and team defensive rating rank. This confirms that voters heavily reward defensive impact within strong team contexts.

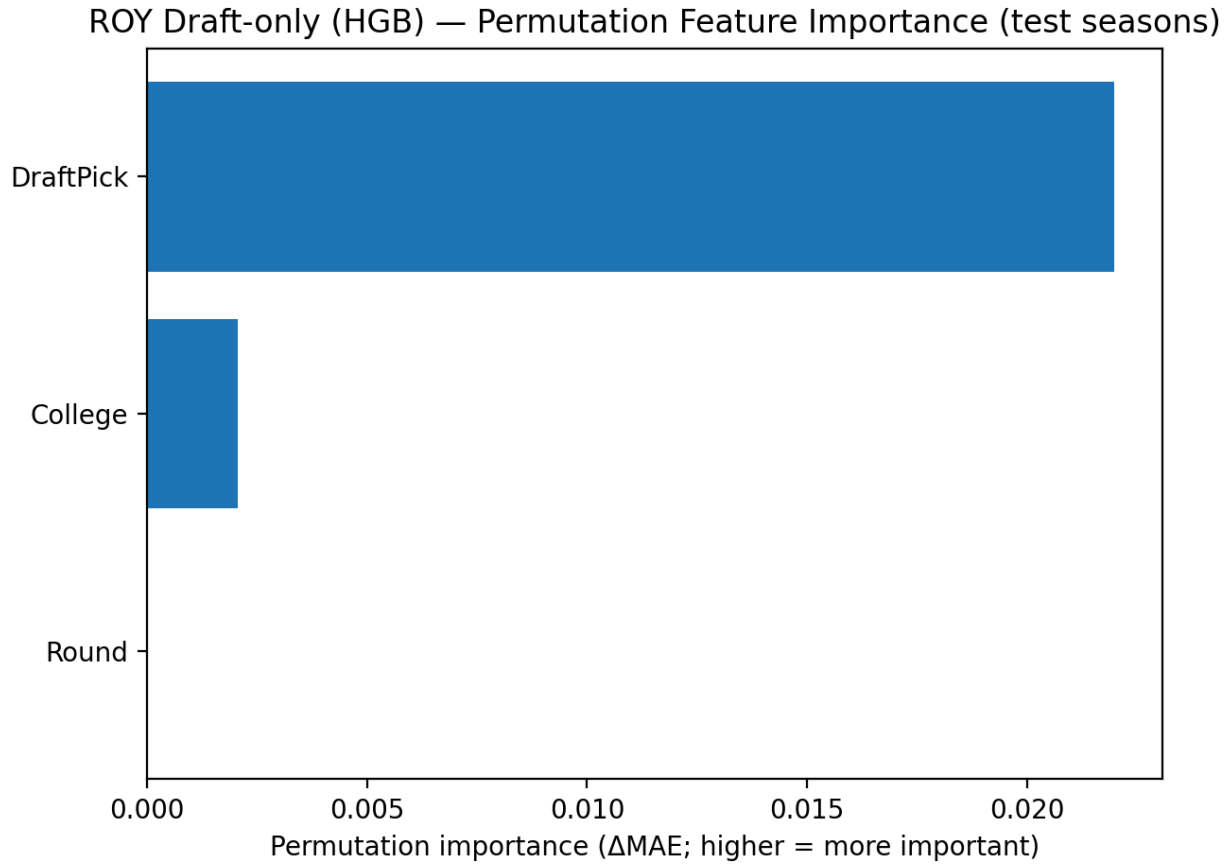
DPOY Projection

Rank	Player	Tm	Predicted_Share	Probability
1.000	Rudy Gobert	MIN	0.274	0.390
2.000	Victor Wembanyama	SAS	0.124	0.177
3.000	Evan Mobley	CLE	0.117	0.166
4.000	Amen Thompson	HOU	0.069	0.099
5.000	Bam Adebayo	MIA	0.051	0.073
6.000	Chet Holmgren	OKC	0.034	0.049
7.000	Scottie Barnes	TOR	0.019	0.027
8.000	Ausar Thompson	DET	0.014	0.020

The model projects Rudy Gobert as the preseason favorite, followed by Victor Wembanyama and Evan Mobley. Notably, team defensive rank remains a major factor in the model's ranking.

ROY Results

Rookie of the Year presents a different modeling challenge because rookies have no NBA history. While an NBA-stat-based rookie model performs well once full-season statistics are available, preseason forecasting must rely on draft information.



The draft-only model uses DraftPick, Round, and College as predictors. DraftPick overwhelmingly dominates, which is consistent with historical trends: higher draft picks receive greater opportunity and media visibility.

Although the draft-only model's Top-1 accuracy in holdout seasons was modest (0.20), this reflects the inherent unpredictability of rookie development rather than model failure.

ROY Projection

Rank	Player	DraftPick	Round	College	Predicted_Share	Probability
1.000	Cooper Flagg	1.000	1.000	Duke	0.449	0.479
2.000	Kon Knueppel	4.000	1.000	Duke	0.195	0.208
3.000	VJ Edgecombe	3.000	1.000	Baylor	0.163	0.174
4.000	Derik Queen	13.000	1.000	Maryland	0.061	0.065
5.000	Cedric Coward	11.000	1.000	Washington State	0.043	0.046
6.000	Egor Dëmin	8.000	1.000	BYU	0.017	0.018
7.000	Collin Murray-Boyles	9.000	1.000	South Carolina	0.007	0.008
8.000	Ryan Kalkbrenner	34.000	2.000	Creighton	0.002	0.002

Cooper Flagg projects as the preseason ROY favorite (≈48% probability), largely driven by the historical success rate of #1 overall picks. Later picks show sharply decreasing probability, reflecting opportunity constraints.

Limitations

These models assume that last-season performance (for MVP/DPOY) and draft position (for ROY) are the strongest available preseason predictors. They do not incorporate injuries, roster changes, or narrative dynamics, which often influence award outcomes. Additionally, defensive metrics remain inherently noisy relative to offensive metrics.

Conclusion

This project demonstrates that NBA award voting patterns are statistically consistent across eras. MVP outcomes are strongly driven by advanced impact metrics and team success. DPOY selections emphasize defensive impact within top-tier team defenses. ROY outcomes are heavily influenced by draft position, reflecting opportunity and visibility.

While narrative always plays a role in awards voting, the statistical backbone of MVP and DPOY selection is surprisingly stable. When evaluated using proper time-based validation, ensemble models reliably identify likely winners using only historical data.